

MODULE 5

1. Capture and analyze ARP packets using Wireshark. Inspect the ARP request and reply frames, and discuss the role of the sender's IP and MAC address in these packets.

The ARP request is a broadcast frame sent to all devices in the network.

- Destination MAC: FF:FF:FF:FF:FF:FF
- Sender IP: Host IP address
- Sender MAC: Host MAC address
- Target IP: IP address being resolved
- Target MAC: Unknown (00:00:00:00:00:00)

The ARP reply is a unicast frame sent back to the requester.

- Destination MAC: Requester's MAC
- Sender IP: Target IP
- Sender MAC: Target device MAC
- Target IP: Requester IP
- Target MAC: Requester MAC

Role of Sender IP and MAC

- The Sender IP address provides logical identification and ensures the reply is sent to the correct IP host.
- The Sender MAC address provides physical identification required for frame delivery in the local network.

30766	-482.7119944...	Intel_e3:e5:09	Broadcast	ARP	42 Who has 10.13.242.36? Tell 10.13.242.153
30767	-482.7004118...	1a:37:0b:1e:6a:cd	Intel_e3:e5:09	ARP	42 10.13.242.36 is at 1a:37:0b:1e:6a:cd

ARP cache on the laptop

```
Select Administrator: Windows PowerShell
Install the latest PowerShell for new features and improvements! h

PS C:\WINDOWS\system32> arp -a

Interface: 10.13.242.153 --- 0xc
Internet Address      Physical Address      Type
10.13.242.36          1a-37-0b-1e-6a-cd     dynamic
10.13.242.255         ff-ff-ff-ff-ff-ff     static
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.251           01-00-5e-00-00-fb     static
224.0.0.252           01-00-5e-00-00-fc     static
239.255.255.250       01-00-5e-7f-ff-fa     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 192.168.56.1 --- 0x11
Internet Address      Physical Address      Type
192.168.56.255        ff-ff-ff-ff-ff-ff     static
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.251           01-00-5e-00-00-fb     static
224.0.0.252           01-00-5e-00-00-fc     static
239.255.255.250       01-00-5e-7f-ff-fa     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static

PS C:\WINDOWS\system32> arp -d *
PS C:\WINDOWS\system32> arp -a

Interface: 10.13.242.153 --- 0xc
Internet Address      Physical Address      Type
224.0.0.22            01-00-5e-00-00-16     static

Interface: 192.168.56.1 --- 0x11
Internet Address      Physical Address      Type
224.0.0.22            01-00-5e-00-00-16     static

PS C:\WINDOWS\system32> ping 10.13.242.36

Pinging 10.13.242.36 with 32 bytes of data:
Reply from 10.13.242.36: bytes=32 time=9ms TTL=64
Reply from 10.13.242.36: bytes=32 time=10ms TTL=64
Reply from 10.13.242.36: bytes=32 time=22ms TTL=64
Reply from 10.13.242.36: bytes=32 time=2ms TTL=64

Ping statistics for 10.13.242.36:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 22ms, Average = 10ms
PS C:\WINDOWS\system32>
```

ARP Request

```
Wireshark - Packet 30766 - WiFi

▶ Frame 30766: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{8D44E5C3-21A6-415F-A343-6B9E6FBC225F}, id 0
▼ Ethernet II, Src: Intel_e3:e5:09 (b0:3c:dc:e3:e5:09), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
  ▼ Destination: Broadcast (ff:ff:ff:ff:ff:ff)
    .... ..1. .... = LG bit: Locally administered address (this is NOT the factory default)
    .... ..1. .... = IG bit: Group address (multicast/broadcast)
  ▼ Source: Intel_e3:e5:09 (b0:3c:dc:e3:e5:09)
    .... ..0. .... = LG bit: Globally unique address (factory default)
    .... ..0. .... = IG bit: Individual address (unicast)
  Type: ARP (0x0806)
  [Stream index: 1]
▼ Address Resolution Protocol (request)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: request (1)
  Sender MAC address: Intel_e3:e5:09 (b0:3c:dc:e3:e5:09)
  Sender IP address: 10.13.242.153 (10.13.242.153)
  Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)
  Target IP address: 10.13.242.36 (10.13.242.36)

0000  ff ff ff ff ff ff b0 3c dc e3 e5 09 08 06 00 01  ....-<-.....
0010  08 00 06 04 00 01 b0 3c dc e3 e5 09 0a 0d f2 99  ....<.....
0020  00 00 00 00 00 00 0a 0d f2 24  ....-$
```

ARP Response

```
Wireshark - Packet 30767 - WiFi

▶ Frame 30767: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{8D44E5C3-21A6-415F-A343-6B9E6FBC225F}, id 0
▼ Ethernet II, Src: 1a:37:0b:1e:6a:cd (1a:37:0b:1e:6a:cd), Dst: Intel_e3:e5:09 (b0:3c:dc:e3:e5:09)
  ▼ Destination: Intel_e3:e5:09 (b0:3c:dc:e3:e5:09)
    .... ..0. .... = LG bit: Globally unique address (factory default)
    .... ..0. .... = IG bit: Individual address (unicast)
  ▼ Source: 1a:37:0b:1e:6a:cd (1a:37:0b:1e:6a:cd)
    .... ..1. .... = LG bit: Locally administered address (this is NOT the factory default)
    .... ..0. .... = IG bit: Individual address (unicast)
  Type: ARP (0x0806)
  [Stream index: 0]
▼ Address Resolution Protocol (reply)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: reply (2)
  Sender MAC address: 1a:37:0b:1e:6a:cd (1a:37:0b:1e:6a:cd)
  Sender IP address: 10.13.242.36 (10.13.242.36)
  Target MAC address: Intel_e3:e5:09 (b0:3c:dc:e3:e5:09)
  Target IP address: 10.13.242.153 (10.13.242.153)

0000  b0 3c dc e3 e5 09 1a 37 0b 1e 6a cd 08 06 00 01  ..<.....7..j.....
0010  08 00 06 04 00 02 1a 37 0b 1e 6a cd 0a 0d f2 24  ....7..j.....$
0020  b0 3c dc e3 e5 09 0a 0d f2 99  ..<.....
```

2. Using Packet Tracer, simulate an ARP spoofing attack. Analyze the behavior of devices on the network when they receive a malicious ARP response.

ARP spoofing cannot be practically performed in Cisco Packet Tracer because the tool does not support crafting or sending malicious ARP packets. It also does not allow manual modification of ARP tables using commands like `arp -s`. Therefore, real attack simulation is not possible within this environment.

ARP Spoofing:

- ARP (Address Resolution Protocol) is used to map an IP address to a MAC address within a local network. ARP spoofing is an attack in which a malicious device sends fake ARP replies to other devices on the network.
- ARP is a stateless protocol and does not verify the authenticity of ARP replies. Devices accept ARP responses without validation, making the protocol vulnerable to spoofing attacks.

During ARP SPOOFING

- The attacker sends a forged ARP reply to a victim device.
- This reply falsely associates the attacker's MAC address with the IP address of another device which is usually the default gateway.
- The victim updates its ARP table with this incorrect mapping.
- Traffic intended for the legitimate device is redirected to the attacker.

Effects on the Network

- Traffic gets misdirected to the attacker
- Possibility of Man-in-the-Middle (MITM) attacks
- Data interception or modification
- Packet loss or denial of service

3. Manually configure static IPs on the client devices (like Pc or your mobile phone) and verify connectivity using ping.

```
yasha@yasha-VirtualBox: ~  
yasha@yasha-VirtualBox:~$ sudo ip addr flush dev enp0s8  
yasha@yasha-VirtualBox:~$ ip link show enp0s8  
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFA  
ULT group default qlen 1000  
    link/ether 08:00:27:fe:9f:8e brd ff:ff:ff:ff:ff:ff
```

Statically assigning an ip to the interface

```
yasha@yasha-VirtualBox:~$ sudo ip addr add 192.168.56.10/24 dev enp0s8  
yasha@yasha-VirtualBox:~$ sudo ip link set enp0s8 up  
yasha@yasha-VirtualBox:~$ ip addr show enp0s8  
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group def  
ault qlen 1000  
    link/ether 08:00:27:fe:9f:8e brd ff:ff:ff:ff:ff:ff  
    inet 192.168.56.101/24 brd 192.168.56.255 scope global dynamic noprefixroute enp0s8  
        valid_lft 240sec preferred_lft 240sec  
    inet 192.168.56.10/24 scope global secondary enp0s8  
        valid_lft forever preferred_lft forever  
    inet6 fe80::a2a2:6e80:9b0d:3784/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
yasha@yasha-VirtualBox:~$
```

Pinging to check the connectivity

```
yasha@yasha-VirtualBox:~$ ping -c 3 192.168.56.1  
PING 192.168.56.1 (192.168.56.1) 56(84) bytes of data.  
64 bytes from 192.168.56.1: icmp_seq=1 ttl=255 time=3.22 ms  
64 bytes from 192.168.56.1: icmp_seq=2 ttl=255 time=4.15 ms  
64 bytes from 192.168.56.1: icmp_seq=3 ttl=255 time=4.03 ms  
  
--- 192.168.56.1 ping statistics ---  
3 packets transmitted, 3 received, 0% packet loss, time 2003ms  
rtt min/avg/max/mdev = 3.222/3.800/4.146/0.411 ms
```

Communicating between windows and linux vm using the VirtualBox host only adapter

```
C:\Users\yasha>ping 192.168.56.10

Pinging 192.168.56.10 with 32 bytes of data:
Request timed out.
Reply from 192.168.56.10: bytes=32 time=1ms TTL=64
Reply from 192.168.56.10: bytes=32 time=1ms TTL=64
Reply from 192.168.56.10: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.56.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\Users\yasha>ping 192.168.56.10

Pinging 192.168.56.10 with 32 bytes of data:
Reply from 192.168.56.10: bytes=32 time=2ms TTL=64
Reply from 192.168.56.10: bytes=32 time=2ms TTL=64
Reply from 192.168.56.10: bytes=32 time=2ms TTL=64
Reply from 192.168.56.10: bytes=32 time=4ms TTL=64

Ping statistics for 192.168.56.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 4ms, Average = 2ms

C:\Users\yasha>
```

- To configure a static IP address on the Linux virtual machine use `ip addr add` command.
- The interface `enp0s8` belongs to the VirtualBox host-only network and normally receives the IP address `192.168.56.101` dynamically from the DHCP server provided by the host.
- When the command `sudo ip addr add 192.168.56.10/24 dev enp0s8` is executed, a secondary static IP address is assigned to the interface while the original DHCP address still remains.
- The command `ip addr flush dev enp0s8` removes all IP addresses from that interface.
- If the default DHCP configuration is required again, the manually added address can be removed using `sudo ip addr del 192.168.56.10/24 dev enp0s8` or the network service can be restarted so that the DHCP client reassigns the default address.
- Connectivity between the host and the VM can then be verified using the `ping` command.

4. Use Wireshark to capture DHCP Discover, Offer, Request, and Acknowledge messages and explain the process.

To generate DHCP traffic, the system IP was released and renewed using:

- `ipconfig /release`
- `ipconfig /renew`

DORA PROCESS

435727	7661.2266673...	0.0.0.0	255.255.255.255	DHCP	342 DHCP Discover	- Transaction ID 0xad5270df
435728	7661.2384180...	10.13.242.36	10.13.242.153	DHCP	352 DHCP Offer	- Transaction ID 0xad5270df
435729	7661.2483262...	0.0.0.0	255.255.255.255	DHCP	370 DHCP Request	- Transaction ID 0xad5270df
435730	7661.2594025...	10.13.242.36	10.13.242.153	DHCP	352 DHCP ACK	- Transaction ID 0xad5270df

Dhcp discover

```
Wireshark - Packet 435727 - Wifi
  Frame 435727: Packet, 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface \Device\NPF_{8D44E5C3-21A6-415F-A343-6B9E6FBC225F}, id 0
  Ethernet II, Src: Intel_e3:e5:09 (b0:3c:dc:e3:e5:09), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
    Destination: Broadcast (ff:ff:ff:ff:ff:ff)
    Source: a00b571c-02f7-4ed2-99bf-1976cb06467b.local (b0:3c:dc:e3:e5:09)
    Type: IPv4 (0x0800)
    [Stream index: 1]
  Internet Protocol Version 4, Src: 0.0.0.0 (0.0.0.0), Dst: 255.255.255.255 (255.255.255.255)
  User Datagram Protocol, Src Port: bootpc (68), Dst Port: bootps (67)
  Dynamic Host Configuration Protocol (Discover)
    Message type: Boot Request (1)
    Hardware type: Ethernet (0x01)
    Hardware address length: 6
    Hops: 0
    Transaction ID: 0xad5270df
    Seconds elapsed: 0
    Bootp flags: 0x0000 (Unicast)
    Client IP address: ab289198-13be-40e2-a0ed-21a48d61544e.local (0.0.0.0)
    Your (client) IP address: ab289198-13be-40e2-a0ed-21a48d61544e.local (0.0.0.0)
    Next server IP address: ab289198-13be-40e2-a0ed-21a48d61544e.local (0.0.0.0)
    Relay agent IP address: ab289198-13be-40e2-a0ed-21a48d61544e.local (0.0.0.0)
    Client MAC address: a00b571c-02f7-4ed2-99bf-1976cb06467b.local (b0:3c:dc:e3:e5:09)
    Client hardware address padding: 00000000000000000000
    Server host name not given
    Boot file name not given
    Magic cookie: DHCP
    Option: (53) DHCP Message Type (Discover)
    Option: (61) Client identifier
    Option: (12) Host Name
    Option: (60) Vendor class identifier
    Option: (55) Parameter Request List
    Option: (255) End
    Padding: 00000000

0020  ff ff 00 44 00 43 01 34  8c 20 01 01 06 00 ad 52  b0 C-4 + .... R
0030  70 df 00 00 00 00 00 00  00 00 00 00 00 00 00 00  p.....
0040  00 00 00 00 00 00 b0 3c  dc e3 e5 09 00 00 00 00  .....< .....
0050  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00  .....
```

Dhcp offer

```
Wireshark - Packet 435728 - WiFi
  Frame 435728: Packet, 352 bytes on wire (2816 bits), 352 bytes captured (2816 bits) on interface \Device\NPF_{8D44E5C3-21A6-415F-A343-6B9E6FBC225F}, id 0
  Ethernet II, Src: 32:e4:7f:3d:ea:1d (32:e4:7f:3d:ea:1d), Dst: a00b571c-02f7-4ed2-99bf-1976cb06467b.local (b0:3c:dc:e3:e5:09)
  Internet Protocol Version 4, Src: 10.13.242.36 (10.13.242.36), Dst: a00b571c-02f7-4ed2-99bf-1976cb06467b.local (10.13.242.153)
  User Datagram Protocol, Src Port: bootps (67), Dst Port: bootpc (68)
  Dynamic Host Configuration Protocol (Offer)
    Message type: Boot Reply (2)
    Hardware type: Ethernet (0x01)
    Hardware address length: 6
    Hops: 0
    Transaction ID: 0xad5270df
    Seconds elapsed: 0
    Bootp flags: 0x0000 (Unicast)
    Client IP address: ab289198-13be-40e2-a0ed-21a48d61544e.local (0.0.0.0)
    Your (client) IP address: a00b571c-02f7-4ed2-99bf-1976cb06467b.local (10.13.242.153)
    Next server IP address: 10.13.242.36 (10.13.242.36)
    Relay agent IP address: ab289198-13be-40e2-a0ed-21a48d61544e.local (0.0.0.0)
    Client MAC address: a00b571c-02f7-4ed2-99bf-1976cb06467b.local (b0:3c:dc:e3:e5:09)
    Client hardware address padding: 00000000000000000000
    Server host name not given
    Boot file name not given
    Magic cookie: DHCP
    Option: (53) DHCP Message Type (Offer)
    Option: (54) DHCP Server Identifier (10.13.242.36)
    Option: (51) IP Address Lease Time
    Option: (58) Renewal Time Value
      Length: 4
      Renewal Time Value: 29 minutes, 59 seconds (1799)
    Option: (59) Rebinding Time Value
    Option: (1) Subnet Mask (255.255.255.0)
    Option: (28) Broadcast Address (10.13.242.255)
    Option: (3) Router
    Option: (6) Domain Name Server
      Length: 4
      Domain Name Server: 10.13.242.36 (10.13.242.36)
    Option: (43) Vendor-Specific Information
    Option: (255) End
    Padding: 00
```

Dhcp request

```
Wireshark - Packet 435729 - WiFi
  Frame 435729: Packet, 370 bytes on wire (2960 bits), 370 bytes captured (2960 bits) on interface \Device\NPF_{8D44E5C3-21A6-415F-A343-6B9E6FBC225F}, id 1
  Ethernet II, Src: a00b571c-02f7-4ed2-99bf-1976cb06467b.local (b0:3c:dc:e3:e5:09), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
  Destination: Broadcast (ff:ff:ff:ff:ff:ff)
    ....1. .... = IG bit: Locally administered address (this is NOT the factory c
    ....1. .... = IG bit: Group address (multicast/broadcast)
  Source: a00b571c-02f7-4ed2-99bf-1976cb06467b.local (b0:3c:dc:e3:e5:09)
    ....0. .... = LG bit: Globally unique address (factory default)
    ....0. .... = IG bit: Individual address (unicast)
  Type: IPv4 (0x0800)
  [Stream index: 1]
  Internet Protocol Version 4, Src: ab289198-13be-40e2-a0ed-21a48d61544e.local (0.0.0.0), Dst: 255.255.255.255
  User Datagram Protocol, Src Port: bootpc (68), Dst Port: bootps (67)
    Source Port: bootpc (68)
    Destination Port: bootps (67)
    Length: 336
    Checksum: 0xd8d7 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 366]
    [Stream Packet Number: 5]
    [Timestamps]
  UDP payload (328 bytes)
  Dynamic Host Configuration Protocol (Request)
    Message type: Boot Request (1)
    Hardware type: Ethernet (0x01)
    Hardware address length: 6
    Hops: 0
    Transaction ID: 0xad5270df
    Seconds elapsed: 0
    Bootp flags: 0x0000 (Unicast)
    Client IP address: ab289198-13be-40e2-a0ed-21a48d61544e.local (0.0.0.0)
    Your (client) IP address: ab289198-13be-40e2-a0ed-21a48d61544e.local (0.0.0.0)
    Next server IP address: ab289198-13be-40e2-a0ed-21a48d61544e.local (0.0.0.0)
    Relay agent IP address: ab289198-13be-40e2-a0ed-21a48d61544e.local (0.0.0.0)
    Client MAC address: a00b571c-02f7-4ed2-99bf-1976cb06467b.local (b0:3c:dc:e3:e5:09)
    Client hardware address padding: 00000000000000000000
    Server host name not given
    Boot file name not given
    Magic cookie: DHCP
    Option: (53) DHCP Message Type (Request)
      Length: 1
      DHCP: Request (3)
```


Contd

```
▼ Option: (61) Client identifier
  Length: 7
  Hardware type: Ethernet (0x01)
  Client MAC address: a00b571c-02f7-4ed2-99bf-1976cb06467b.local (b0:3c:dc:e3:e5:09)
▼ Option: (50) Requested IP Address (10.13.242.153)
  Length: 4
  Requested IP Address: a00b571c-02f7-4ed2-99bf-1976cb06467b.local (10.13.242.153)
▼ Option: (54) DHCP Server Identifier (10.13.242.36)
  Length: 4
  DHCP Server Identifier: 10.13.242.36 (10.13.242.36)
▼ Option: (12) Host Name
  Length: 15
  Host Name: LAPTOP-5LJ7FR20
▼ Option: (81) Client Fully Qualified Domain Name
  Length: 18
  ▶ Flags: 0x00
  A-RR result: 0
  PTR-RR result: 0
  Client name: LAPTOP-5LJ7FR20
▼ Option: (60) Vendor class identifier
  Length: 8
  Vendor class identifier: MSFT 5.0
▼ Option: (55) Parameter Request List
  Length: 14
  Parameter Request List Item: (1) Subnet Mask
  Parameter Request List Item: (3) Router
  Parameter Request List Item: (6) Domain Name Server
  Parameter Request List Item: (15) Domain Name
  Parameter Request List Item: (31) Perform Router Discover
  Parameter Request List Item: (33) Static Route
  Parameter Request List Item: (43) Vendor-Specific Information
  Parameter Request List Item: (44) NetBIOS over TCP/IP Name Server
  Parameter Request List Item: (46) NetBIOS over TCP/IP Node Type
  Parameter Request List Item: (47) NetBIOS over TCP/IP Scope
  Parameter Request List Item: (119) Domain Search
  Parameter Request List Item: (121) Classless Static Route
  Parameter Request List Item: (249) Private/Classless Static Route (Microsoft)
  Parameter Request List Item: (252) Private/Proxy autodiscovery
▼ Option: (255) End
  Option End: 255
```

Dhcp accept

```
Wireshark - Packet 435730 - WiFi
  Frame 435730: Packet, 352 bytes on wire (2816 bits), 352 bytes captured (2816 bits) on interface \Device\NPF...
  Ethernet II, Src: 32:e4:7f:3d:ea:1d (32:e4:7f:3d:ea:1d), Dst: a00b571c-02f7-4ed2-99bf-1976cb06467b
  Internet Protocol Version 4, Src: 10.13.242.36 (10.13.242.36), Dst: a00b571c-02f7-4ed2-99bf-1976cb06467b
  User Datagram Protocol, Src Port: bootps (67), Dst Port: bootpc (68)
  Dynamic Host Configuration Protocol (ACK)
    Message type: Boot Reply (2)
    Hardware type: Ethernet (0x01)
    Hardware address length: 6
    Hops: 0
    Transaction ID: 0xad5270df
    Seconds elapsed: 0
  Bootp flags: 0x0000 (Unicast)
    Client IP address: ab289198-13be-40e2-a0ed-21a48d61544e.local (0.0.0.0)
    Your (client) IP address: a00b571c-02f7-4ed2-99bf-1976cb06467b.local (10.13.242.153)
    Next server IP address: 10.13.242.36 (10.13.242.36)
    Relay agent IP address: ab289198-13be-40e2-a0ed-21a48d61544e.local (0.0.0.0)
    Client MAC address: a00b571c-02f7-4ed2-99bf-1976cb06467b.local (b0:3c:dc:e3:e5:09)
    Client hardware address padding: 00000000000000000000
    Server host name not given
    Boot file name not given
    Magic cookie: DHCP
  Option: (53) DHCP Message Type (ACK)
    Length: 1
    DHCP: ACK (5)
  Option: (54) DHCP Server Identifier (10.13.242.36)
    Length: 4
    DHCP Server Identifier: 10.13.242.36 (10.13.242.36)
  Option: (51) IP Address Lease Time
    Length: 4
    IP Address Lease Time: 59 minutes, 59 seconds (3599)
  Option: (58) Renewal Time Value
    Length: 4
    Renewal Time Value: 29 minutes, 59 seconds (1799)
  Option: (59) Rebinding Time Value
    Length: 4
    Rebinding Time Value: 52 minutes, 29 seconds (3149)
  Option: (1) Subnet Mask (255.255.255.0)
    Length: 4
    Subnet Mask: 255.255.255.0
  Option: (28) Broadcast Address (10.13.242.255)
    Length: 4
    Broadcast Address: 10.13.242.255 (10.13.242.255)
  Option: (3) Router
    Length: 4
    Router: 10.13.242.36 (10.13.242.36)
  Option: (6) Domain Name Server
    Length: 4
    Domain Name Server: 10.13.242.36 (10.13.242.36)
  Option: (43) Vendor-Specific Information
    Length: 15
    Value: 414e44524f49445f4d455445524544
  Option: (255) End
    Option End: 255
    Padding: 00
```

Contd

```
Wireshark - Packet 435730 - WiFi
  Your (client) IP address: a00b571c-02f7-4ed2-99bf-1976cb06467b.local (10.13.242.153)
  Next server IP address: 10.13.242.36 (10.13.242.36)
  Relay agent IP address: ab289198-13be-40e2-a0ed-21a48d61544e.local (0.0.0.0)
  Client MAC address: a00b571c-02f7-4ed2-99bf-1976cb06467b.local (b0:3c:dc:e3:e5:09)
  Client hardware address padding: 00000000000000000000
  Server host name not given
  Boot file name not given
  Magic cookie: DHCP
  Option: (53) DHCP Message Type (ACK)
    Length: 1
    DHCP: ACK (5)
  Option: (54) DHCP Server Identifier (10.13.242.36)
    Length: 4
    DHCP Server Identifier: 10.13.242.36 (10.13.242.36)
  Option: (51) IP Address Lease Time
    Length: 4
    IP Address Lease Time: 59 minutes, 59 seconds (3599)
  Option: (58) Renewal Time Value
    Length: 4
    Renewal Time Value: 29 minutes, 59 seconds (1799)
  Option: (59) Rebinding Time Value
    Length: 4
    Rebinding Time Value: 52 minutes, 29 seconds (3149)
  Option: (1) Subnet Mask (255.255.255.0)
    Length: 4
    Subnet Mask: 255.255.255.0
  Option: (28) Broadcast Address (10.13.242.255)
    Length: 4
    Broadcast Address: 10.13.242.255 (10.13.242.255)
  Option: (3) Router
    Length: 4
    Router: 10.13.242.36 (10.13.242.36)
  Option: (6) Domain Name Server
    Length: 4
    Domain Name Server: 10.13.242.36 (10.13.242.36)
  Option: (43) Vendor-Specific Information
    Length: 15
    Value: 414e44524f49445f4d455445524544
  Option: (255) End
    Option End: 255
    Padding: 00
```

DHCP Process (DORA)

1. DHCP Discover

The client initiates communication by broadcasting a DHCP Discover message.

- Source IP: 0.0.0.0
- Destination IP: 255.255.255.255
- Purpose: To locate available DHCP servers

2. DHCP Offer

The DHCP server responds with an offer containing an available IP address.

- Contains offered IP address
- Includes configuration parameters

3. DHCP Request

The client broadcasts a request to accept the offered IP address.

- Confirms selected DHCP server
- Requests the offered IP

4. DHCP Acknowledge (ACK)

The server sends an acknowledgment confirming IP allocation.

- Assigns IP address
- Provides subnet mask, gateway, lease time

5. Given an IP address range of 192.168.1.0/24, divide the network into 4 subnets.

Task: Manually calculate the new subnet mask and the range of valid IP addresses for each subnet.

Assign IP addresses from these subnets to devices in Cisco Packet Tracer and verify connectivity using ping between them.

5. IP address range : 192.168.1.0/24

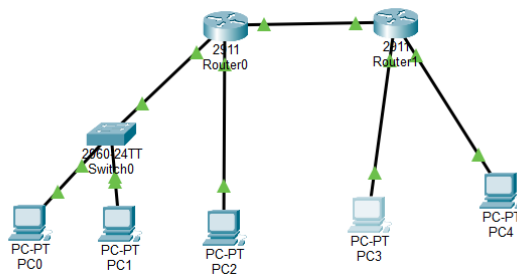
Required subnet : 4

So the subnet mask becomes 192.168.1.0/26

$$\begin{aligned} \text{no. of usable IPs in each subnet} &= 2^6 - 2 \\ &= 64 - 2 \\ &= 62 \end{aligned}$$

Subnet mask : 255.255.255.192

Subnet	Network address	First host	Last host	Broadcast
1	192.168.1.0	192.168.1.1	192.168.1.62	192.168.1.63
2	192.168.1.64	192.168.1.65	192.168.1.126	192.168.1.127
3	192.168.1.128	192.168.1.129	192.168.1.190	192.168.1.191
4	192.168.1.192	192.168.1.193	192.168.1.254	192.168.1.255



Assigning Router0 interfaces

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```

Would you like to enter the initial configuration dialog? [yes/no]:
Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/1
Router(config-if)#ip address 192.168.1.1
% Incomplete command.
Router(config-if)#ip address 192.168.1.1 255.255.255.192
Router(config-if)#no shutdown

Router(config-if)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to down
exit
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to down
end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
  
```

Copy Paste

☐ Top

Router0

Physical Config CLI Attributes

IOS Command Line Interface

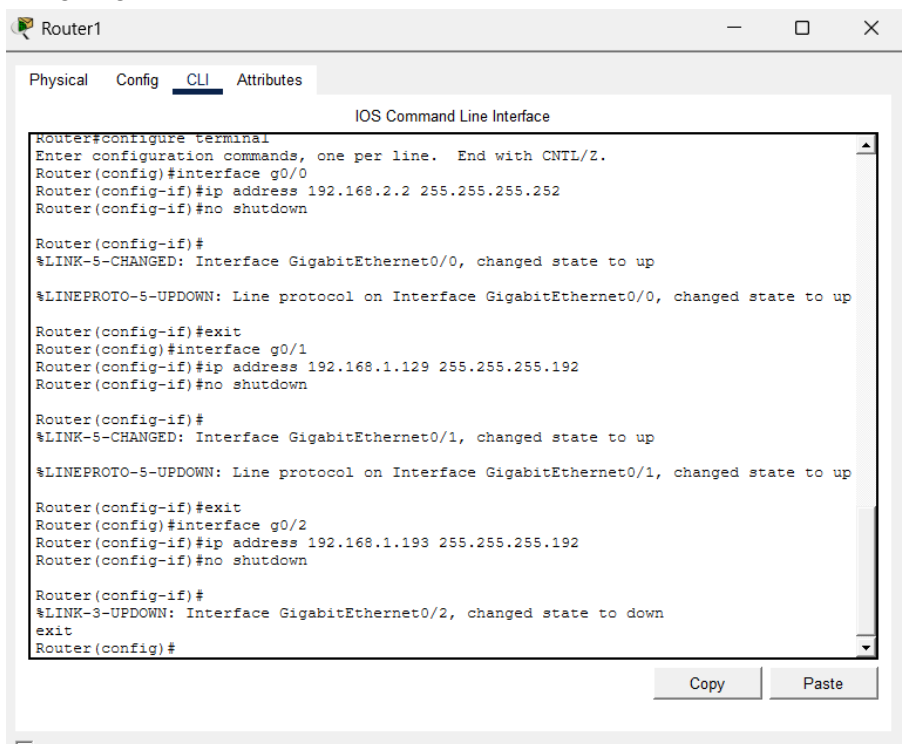
```

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/2
Router(config-if)#ip address 192.168.1.65 255.255.255.192
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
  
```

Assigning Router1 interfaces



Setting up static routes on both the routers

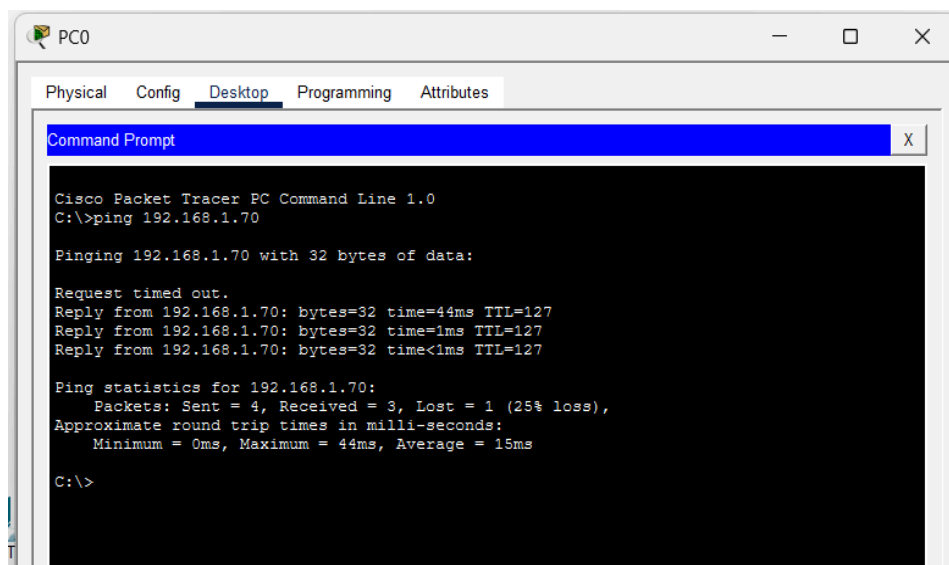
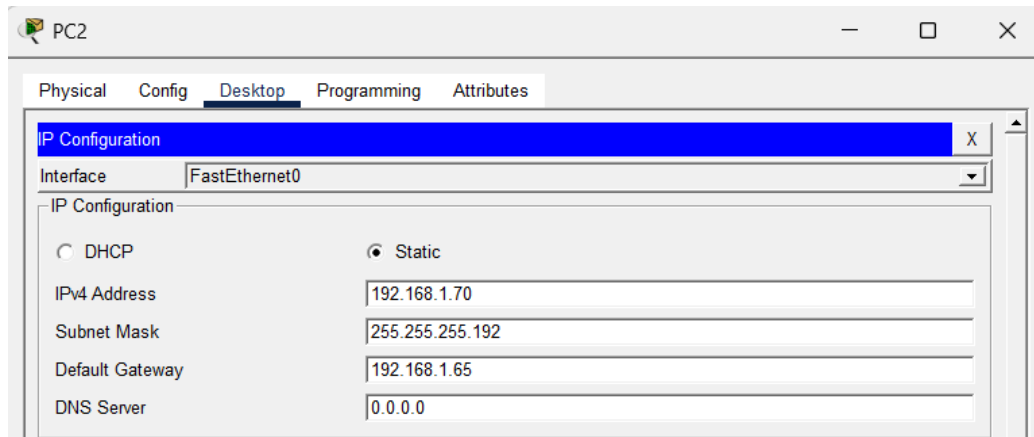
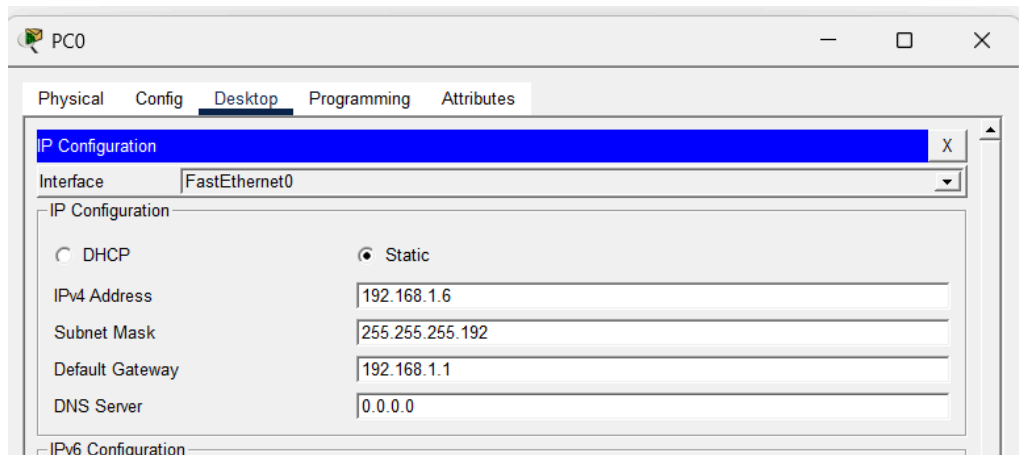
Router0

```
Router(config)#ip route 192.168.1.128 255.255.255.192 192.168.2.2
Router(config)#ip route 192.168.1.192 255.255.255.192 192.168.2.2
Router(config)#
```

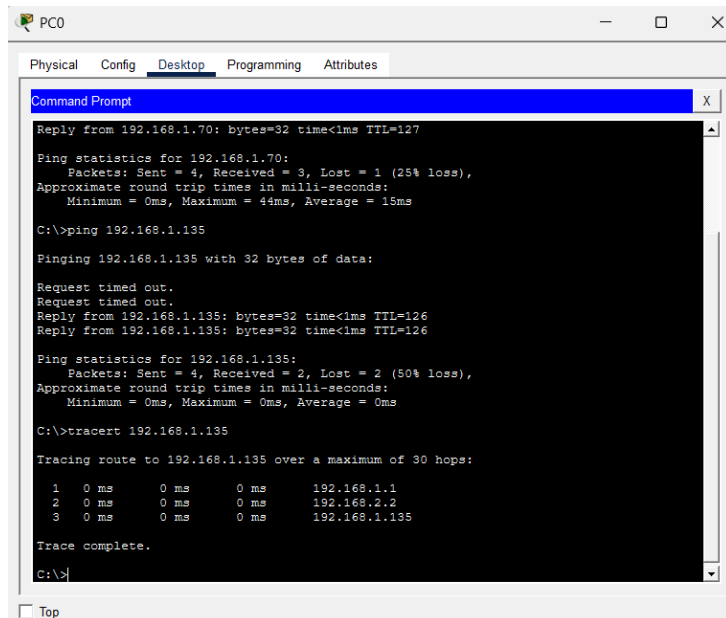
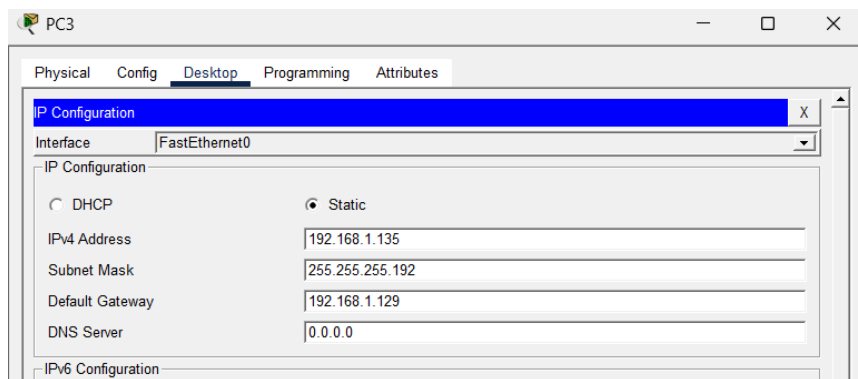
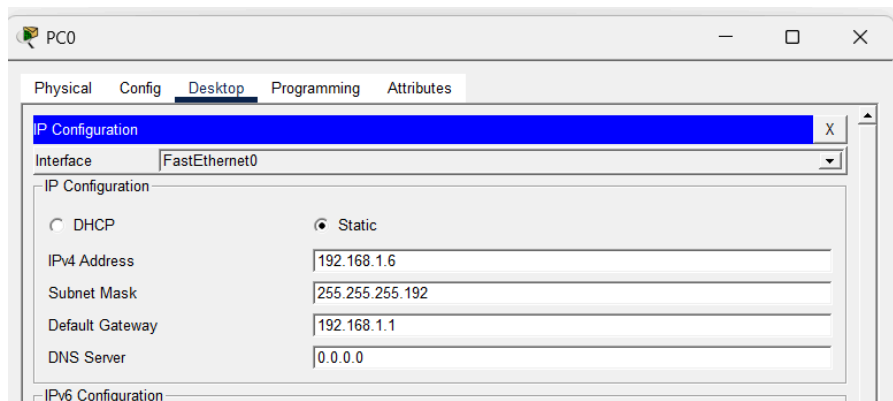
Router1

```
Router(config)#ip route 192.168.1.0 255.255.255.192 192.168.2.1
Router(config)#ip route 192.168.1.64 255.255.255.192 192.168.2.1
Router(config)#
```

Verifying the connectivity between subnet 1 and subnet 2



Verifying the connectivity between subnet 1 and subnet 3 (Default gateway is needed)



6. You are given three IP addresses: 10.1.1.1, 172.16.5.10, and 192.168.1.5.

Task: Identify the class of each IP address (Class A, B, or C). What is the default subnet mask for each class

Provide the range of IP addresses for each class.

10.1.1.1

Class: A

Default Subnet Mask: 255.0.0.0 (/8)

172.16.5.10

Class: B

Default Subnet Mask: 255.255.0.0 (/16)

192.168.1.5

Class: C

Default Subnet Mask: 255.255.255.0 (/24)

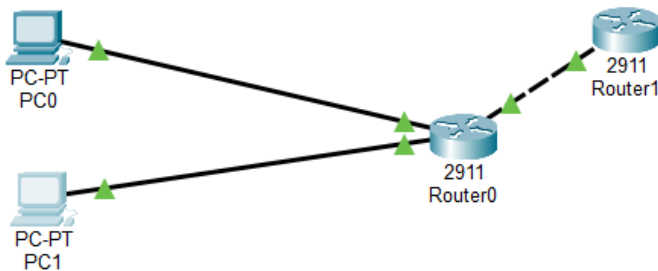
- **Class A:** 1.0.0.0 to 126.255.255.255
- **Class B:** 128.0.0.0 to 191.255.255.255
- **Class C:** 192.0.0.0 to 223.255.255.255

All the given IP addresses belong to private IP ranges as defined by RFC 1918. These addresses are used within local networks and are not routable on the public internet.

7. In Cisco Packet Tracer, create a small network with multiple devices (e.g., 2 PCs and a router). Use private IP addresses (e.g., 192.168.1.x) on the PCs and configure the router to perform NAT to allow the PCs to access the internet.

Task: Test the NAT configuration by pinging an external IP address from the PCs and capture the traffic using Wireshark.

What is the source IP address before and after NAT?



Router 0

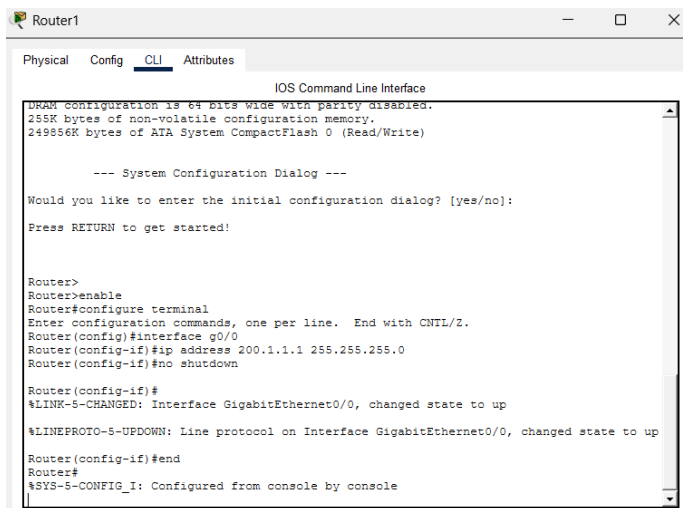
```
Router0
Physical Config CLI Attributes
IOS Command Line Interface
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to
down
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/1
Router(config-if)#ip address 200.1.1.2 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

```

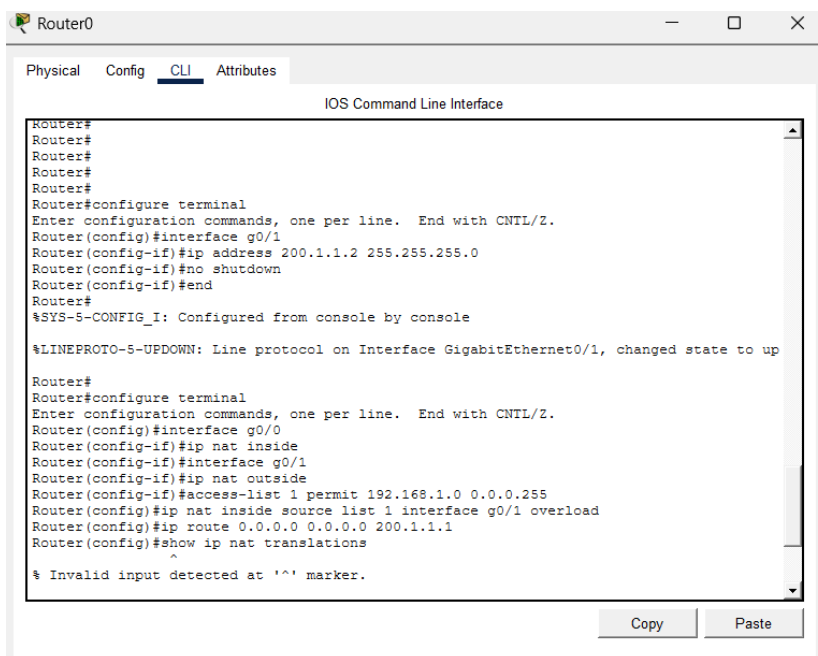
Router>
Router>
Router>enable
Router#configure
Configuring from terminal, memory, or network [terminal]? terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

```

Router 1



NAT configuration



Nat translation

```
Router#show ip nat translations
Pro  Inside global      Inside local      Outside local      Outside global
icmp 200.1.1.2:1       192.168.1.10:1    200.1.1.1:1        200.1.1.1:1
icmp 200.1.1.2:2       192.168.1.10:2    200.1.1.1:2        200.1.1.1:2
icmp 200.1.1.2:3       192.168.1.10:3    200.1.1.1:3        200.1.1.1:3
icmp 200.1.1.2:4       192.168.1.10:4    200.1.1.1:4        200.1.1.1:4

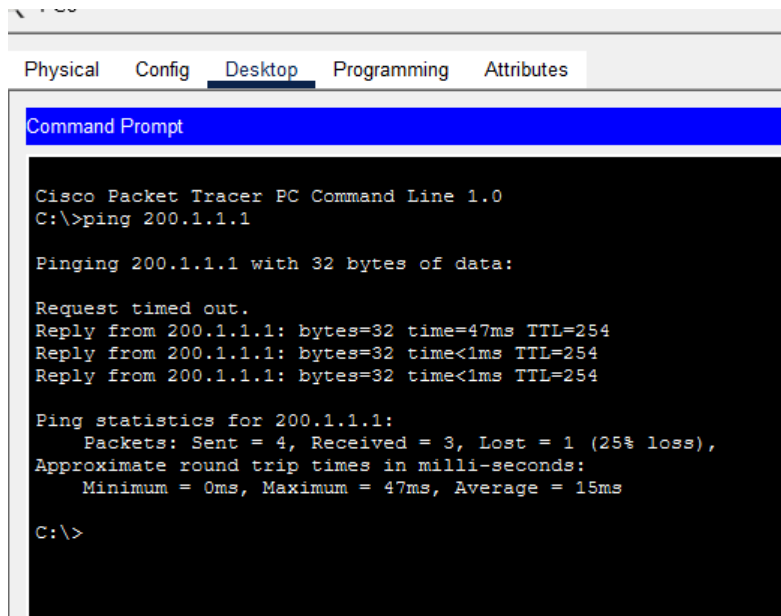
Router#
```

Pinging to verify connectivity

```
Pinging 200.1.1.2 with 32 bytes of data:

Reply from 200.1.1.2: bytes=32 time=1ms TTL=127
Reply from 200.1.1.2: bytes=32 time=1ms TTL=127
Reply from 200.1.1.2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.2: bytes=32 time<1ms TTL=127

Ping statistics for 200.1.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```



```
Physical  Config  Desktop  Programming  Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 200.1.1.1

Pinging 200.1.1.1 with 32 bytes of data:

Request timed out.
Reply from 200.1.1.1: bytes=32 time=47ms TTL=254
Reply from 200.1.1.1: bytes=32 time<1ms TTL=254
Reply from 200.1.1.1: bytes=32 time<1ms TTL=254

Ping statistics for 200.1.1.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 47ms, Average = 15ms

C:\>
```

The PCs were assigned private IP addresses (192.168.1.10 and 192.168.1.20) with a subnet mask of 255.255.255.0 and a default gateway of 192.168.1.1.

The router connected to the internal network was configured with IP address 192.168.1.1 on the inside interface and 200.1.1.2 on the outside interface.

Another router was configured to simulate the internet with IP address 200.1.1.1. Network Address Translation (NAT) was configured on the internal router by designating inside and outside interfaces, creating an access control list to permit the private network, and enabling NAT overload using the command "ip nat inside source list 1 interface g0/1 overload." A default route was also configured to forward traffic to the external network.

The NAT configuration was successfully verified by pinging the external IP address from the PCs and observing address translation using the "show ip nat translations" command.

